INFOSHIELD: Safety-Constrained Information Flow for Trustworthy Multi-Agent LLM Systems

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Abstract

Recent advances in agentic artificial intelligence have transformed large language models (LLMs) from passive conversational systems into autonomous multi-agent systems capable of collaborative reasoning, planning, memory retrieval, tool interaction, and long-horizon task execution. However, these capabilities introduce significant safety risks, including prompt injection, reasoning-state poisoning, unsafe tool execution, memory corruption, and inter-agent propagation of adversarial information. Existing defenses primarily focus on prompt-level filtering or alignment-based fine-tuning, which are insufficient for controlling unsafe information propagation across collaborative multi-agent workflows.

This proposal introduces **INFOSHIELD**, a novel information-theoretic safety framework for autonomous multi-agent LLM systems. Unlike existing information bottleneck defenses that primarily compress adversarial prompts, INFOSHIELD formulates multi-agent safety as a problem of regulating unsafe information propagation across planner, memory, tool-use, retrieval, and inter-agent communication pathways. The framework introduces safety-constrained bottleneck mechanisms that selectively preserve task-relevant collaborative reasoning signals while suppressing adversarial or unsafe information transfer.

INFOSHIELD integrates information bottleneck theory, uncertainty-aware verification, and collaborative reasoning safety mechanisms to mitigate cascading failures in long-horizon autonomous workflows. The proposed framework will be evaluated against prompt injection, shared-memory poisoning, reasoning-state injection, tool-output poisoning, and multi-agent consensus collapse attacks using modern multi-agent frameworks including AutoGen, CrewAI, LangGraph, and OpenAI Agents SDK.

The expected outcome of this research is a theoretically grounded and practically deployable safety architecture for trustworthy multi-agent LLM systems capable of mitigating emerging threats in autonomous collaborative AI environments.

1 Introduction

Large language models have rapidly evolved from passive conversational systems into autonomous multi-agent systems capable of collaborative reasoning, planning, memory retrieval, tool interaction, and long-horizon task execution. Modern frameworks such as AutoGen, CrewAI, LangGraph, and OpenAI Agents SDK enable multiple LLM agents to communicate, coordinate, delegate subtasks, and autonomously solve complex problems with minimal human supervision.

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Despite these advances, multi-agent LLM systems introduce a new class of safety and reliability challenges. In collaborative agentic environments, unsafe or adversarial information may propagate across multiple internal communication pathways, including planner states, shared memory systems, retrieval pipelines, tool outputs, verifier agents, and inter-agent messaging channels. As a result, harmful reasoning signals can spread through the workflow even when the original user prompt appears benign.

Existing safety mechanisms primarily focus on:

- Prompt filtering
- Jailbreak detection
- RLHF-based alignment
- Output moderation
- Adversarial prompt perturbation

Although these methods improve single-model robustness, they fail to address the broader challenge of regulating unsafe information propagation inside collaborative multi-agent reasoning systems. Current defenses rarely model:

- Inter-agent communication safety
- Planner-to-memory information transfer
- Unsafe tool-response propagation
- Long-horizon reasoning contamination
- Multi-agent consensus instability

This research proposes **INFOSHIELD**, a safety-constrained information bottleneck framework for trustworthy multi-agent LLM systems. The central hypothesis is that unsafe behavior in autonomous multi-agent workflows emerges through uncontrolled propagation of adversarial or irrelevant information across planner, memory, tool-use, and inter-agent communication channels. INFOSHIELD aims to regulate these information pathways using bottleneck-based compression and uncertainty-aware verification mechanisms while preserving collaborative reasoning capability and long-horizon task performance.

2 Problem Statement

Current multi-agent LLM systems face several critical vulnerabilities:

- Prompt injection can manipulate downstream collaborative reasoning.
- Tool outputs may introduce adversarial instructions into shared workflows.
- Shared memory systems may accumulate poisoned reasoning traces.
- Inter-agent communication can amplify hallucinations and unsafe decisions.
- Autonomous planners may propagate unsafe latent objectives across agents.

- Existing alignment mechanisms primarily operate at input or output levels rather than internal communication pathways.

In collaborative agentic systems, unsafe information does not remain isolated within a single prompt. Instead, harmful reasoning signals may propagate across planner modules, memory systems, retrieval pipelines, external tools, and communication channels between autonomous agents. This creates cascading safety failures that existing prompt-level defenses cannot adequately mitigate.

Therefore, the core research challenge is not merely detecting harmful prompts, but controlling how unsafe information propagates throughout collaborative reasoning workflows in autonomous multi-agent systems.

> *Can information bottleneck constraints prevent unsafe reasoning propagation across planner, memory, tool, and inter-agent communication channels in autonomous multi-agent LLM systems while preserving reasoning fidelity and long-horizon task performance?*

3 Research Objectives

The primary objective of this research is to develop a safety-constrained information bottleneck framework for trustworthy multi-agent LLM systems.

Specific objectives include:

1. Formulate multi-agent LLM safety as an information propagation control problem.
2. Design bottleneck modules for planner-memory-tool-agent interactions.
3. Develop uncertainty-aware verification mechanisms for collaborative reasoning systems.
4. Introduce attack models targeting inter-agent communication and reasoning-state propagation.
5. Propose metrics for measuring unsafe information transfer across collaborative workflows.
6. Evaluate robustness under adversarial long-horizon multi-agent reasoning scenarios.

4 Research Questions

RQ1: How does unsafe information propagate across planner, memory, tool-use, and inter-agent communication channels in multi-agent LLM workflows?

RQ2: Can information bottleneck constraints suppress adversarial reasoning signals while preserving collaborative task-relevant information?

RQ3: How do shared memory systems and external tool interactions contribute to cascading safety failures in multi-agent environments?

RQ4: Can uncertainty-aware bottlenecking improve robustness against reasoning-state poisoning and inter-agent adversarial propagation?

RQ5: How does information compression affect collaborative reasoning fidelity, communication efficiency, and long-horizon task completion?

5 Proposed Framework: INFOSHIELD

INFOSHIELD introduces safety-aware information bottlenecks across collaborative multi-agent reasoning systems. Instead of compressing only the user prompt, INFOSHIELD regulates information propagation across planner, memory, tool-use, retrieval, and inter-agent communication pathways.

The framework models information flow across:

- User prompt $\rightarrow$ Planner Agent
- Planner Agent $\rightarrow$ Worker Agents
- Worker Agents $\rightarrow$ Tool Interfaces
- Tool Outputs $\rightarrow$ Shared Memory
- Shared Memory $\rightarrow$ Verifier Agents
- Inter-Agent Messages $\rightarrow$ Collaborative Reasoning Layer
- Verifier Agents $\rightarrow$ Final Synthesis Agent

The framework consists of the following components:

1. **Planner Agent:** decomposes complex tasks into collaborative reasoning subtasks.
2. **Worker Agents:** perform specialized reasoning and tool-use operations.
3. **Shared Memory Controller:** stores and retrieves historical reasoning traces and agent interactions.
4. **Information Bottleneck Layer:** compresses intermediate reasoning representations before inter-agent propagation.
5. **Safety Verifier Agents:** estimate uncertainty, adversarial risk, and unsafe propagation likelihood.
6. **Tool Interaction Module:** regulates external API and retrieval interactions.
7. **Final Synthesis Agent:** produces the final response using safety-filtered collaborative reasoning outputs.

6 Mathematical Formulation

Let:

- X denote user input
- Z denote compressed collaborative reasoning representation
- Y denote desired safe output
- M denote shared memory state
- T denote tool outputs
- A denote inter-agent communication states

The objective is to maximize task-relevant collaborative reasoning while minimizing unsafe information propagation:

$$\max I(Z;Y) - \beta I(Z;X_{\text{unsafe}}) \tag{1}$$

where:

- $I(Z;Y)$ preserves collaborative task-relevant reasoning information
- $I(Z;X_{\text{unsafe}})$ penalizes unsafe information propagation
- β controls the safety-compression tradeoff

The overall optimization objective is:

$$\mathcal{L}_{safe} = \mathcal{L}_{task} + \lambda_1 \mathcal{L}_{IB} + \lambda_2 \mathcal{L}_{uncertainty} + \lambda_3 \mathcal{L}_{communication} \tag{2}$$

where:

- $\mathcal{L}_{task}$ measures collaborative task completion performance
- $\mathcal{L}_{IB}$ enforces bottleneck compression
- $\mathcal{L}_{uncertainty}$ penalizes unsafe high-confidence reasoning
- $\mathcal{L}_{communication}$ regulates unsafe inter-agent propagation

7 Novel Contributions

This research aims to make the following contributions:

1. The first formulation of multi-agent LLM safety as an information propagation control problem.
2. A novel safety-constrained information bottleneck architecture for collaborative agentic systems.
3. Introduction of reasoning-state poisoning and inter-agent propagation attacks.
4. Uncertainty-aware bottleneck verification for long-horizon collaborative workflows.
5. New safety metrics for measuring unsafe information propagation across multi-agent communication pathways.
6. Comprehensive benchmarking across AutoGen, CrewAI, LangGraph, and modern multi-agent frameworks.

8 Threat Models

The proposed framework will be evaluated under the following collaborative multi-agent attack settings:

8.1 Prompt Injection

Malicious instructions inserted into user prompts manipulate downstream planner reasoning and collaborative agent coordination.

8.2 Tool Output Poisoning

External APIs and retrieval systems return adversarial content that propagates across shared agent workflows.

8.3 Shared Memory Poisoning

Long-term shared memory systems accumulate unsafe reasoning traces and contaminated agent interactions.

8.4 Reasoning-State Injection

Intermediate collaborative reasoning states are manipulated without directly modifying the original user prompt.

8.5 Inter-Agent Propagation Attacks

Unsafe reasoning signals spread between communicating agents during collaborative task execution.

8.6 Multi-Agent Consensus Collapse

Multiple agents reinforce unsafe or hallucinated reasoning through correlated agreement dynamics.

9 Experimental Setup

9.1 Datasets and Benchmarks

The framework will be evaluated using:

- AdvBench
- EasyJailbreak
- AgentBench
- ToolBench
- HotpotQA
- GSM8K
- MATH

9.2 Frameworks

Experiments will be conducted using:

- AutoGen
- CrewAI
- LangGraph
- OpenAI Agents SDK

9.3 Baselines

INFOSHIELD will be compared against:

- IBProtector
- SmoothLLM
- Semantic Smooth
- RLHF-based safety systems
- Constitutional AI
- Multi-agent verifier frameworks

10 Evaluation Metrics

10.1 Safety Metrics

- Attack Success Rate
- Unsafe Propagation Score
- Inter-Agent Leakage Rate
- Tool Misuse Rate
- Consensus Collapse Rate

10.2 Reasoning Metrics

- Collaborative Task Completion Accuracy
- Reasoning Fidelity
- Hallucination Rate
- Multi-step Consistency

10.3 Efficiency Metrics

- Latency
- Token Overhead
- Communication Overhead
- Memory Consumption

11 Timeline and Task Assignment

Target Deadline: July 21

Phase	Date	Lead	Tasks
Phase 1	Feb–Mar	Elias	Finalize problem formulation and novelty
Phase 2	Mar–Apr	Maleeha	Literature review and related work
Phase 3	Mar–Apr	Mehedi	Multi-agent experimental setup
Phase 4	Apr–May	Elias, Mehedi	Implement INFOSHIELD modules
Phase 5	May	Maleeha, Mehedi	Build attack pipelines
Phase 6	Jun	Mehedi	Run experiments and ablations
Phase 7	Jun	Maleeha	Safety analysis and discussion writing
Phase 8	Jul	Elias	Main paper writing and integration
Phase 9	Jul	All Authors	Final review and polishing
Submission	Jul 21	Elias	Final submission

11.1 Elias Hossain

Lead author responsible for methodology, mathematical formulation, architecture design, experiment supervision, and final submission.

11.2 Maleeha

Responsible for related work, literature review, safety analysis, ethical considerations, and discussion sections.

11.3 Mehedi

Responsible for implementation, experiments, attack pipelines, evaluation metrics, figures, and reproducibility.

12 Expected Outcomes

The expected outcomes include:

1. A new theoretical perspective on collaborative multi-agent safety.
2. A practical safety architecture for autonomous multi-agent LLM systems.
3. Improved robustness against reasoning-state and propagation attacks.
4. New metrics for unsafe collaborative information transfer.
5. A conference-ready paper for AAAI, NeurIPS, or ICLR.

13 Broader Impact

INFOSHIELD may contribute toward safer autonomous AI systems in healthcare, cybersecurity, scientific research, finance, and policy analysis. By regulating unsafe information propagation across collaborative agentic systems, the framework aims to improve trustworthiness and reduce cascading failures in high-stakes AI deployments.

14 Conclusion

This proposal presents INFOSHIELD, a novel information-theoretic framework for trustworthy multi-agent LLM systems. Unlike prompt-level defenses, INFOSHIELD regulates unsafe information propagation across planner, memory, tool-use, retrieval, and inter-agent communication pathways. By combining information bottleneck theory with uncertainty-aware collaborative reasoning verification, the proposed framework aims to establish a new foundation for safe and scalable autonomous multi-agent AI systems.

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